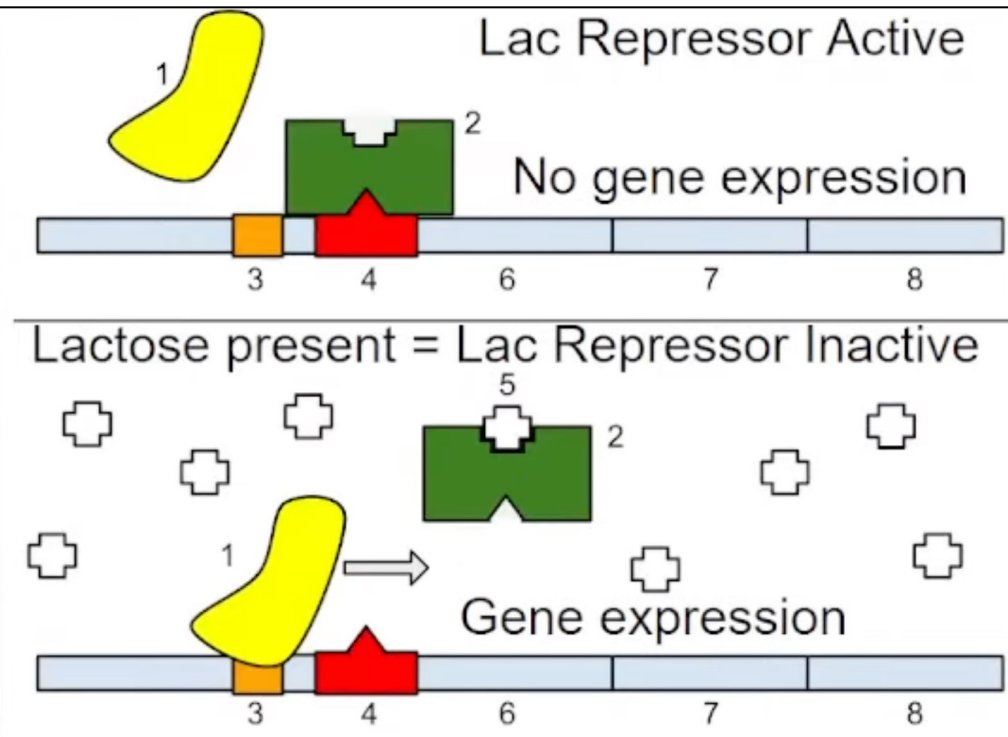
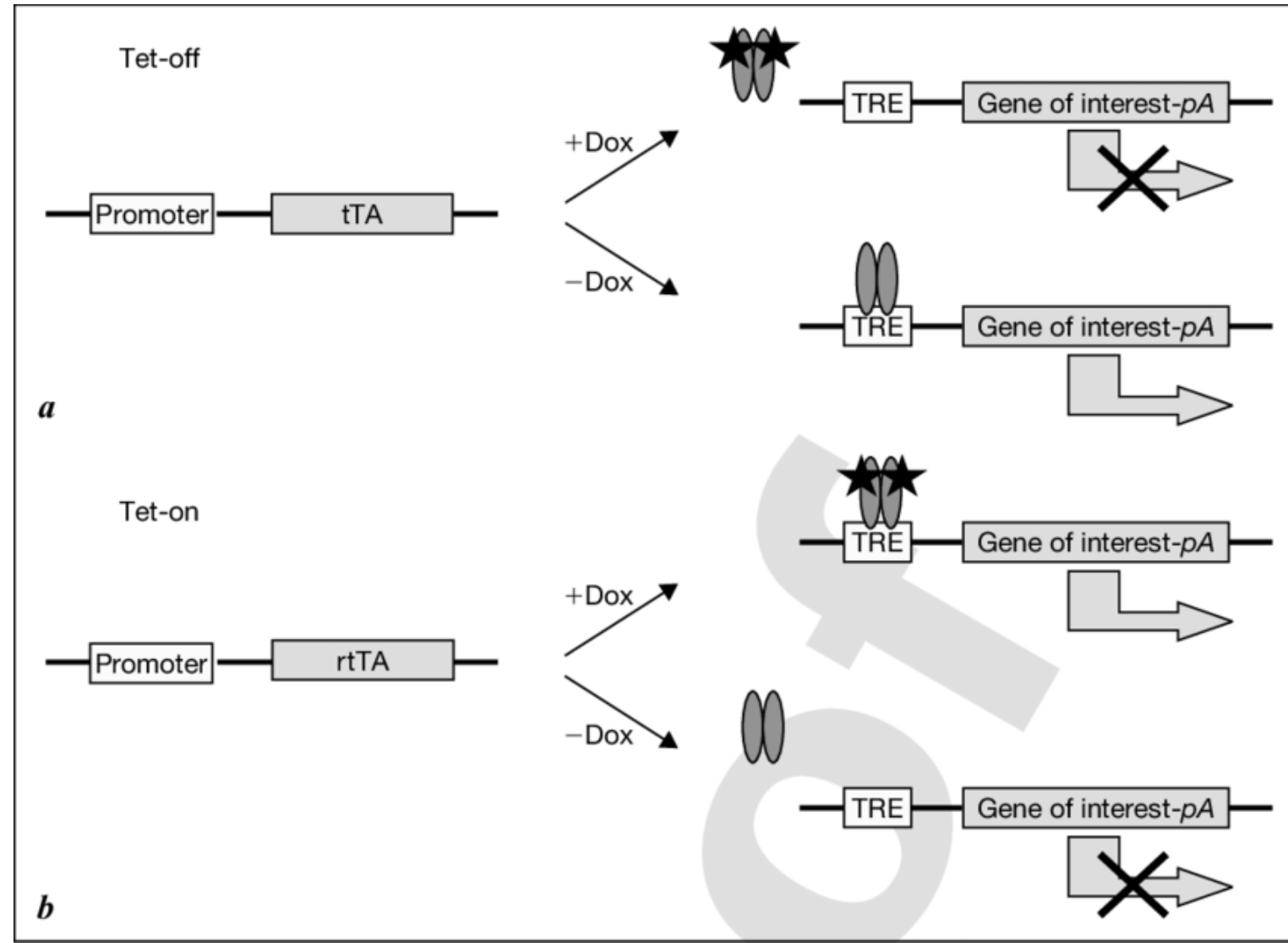


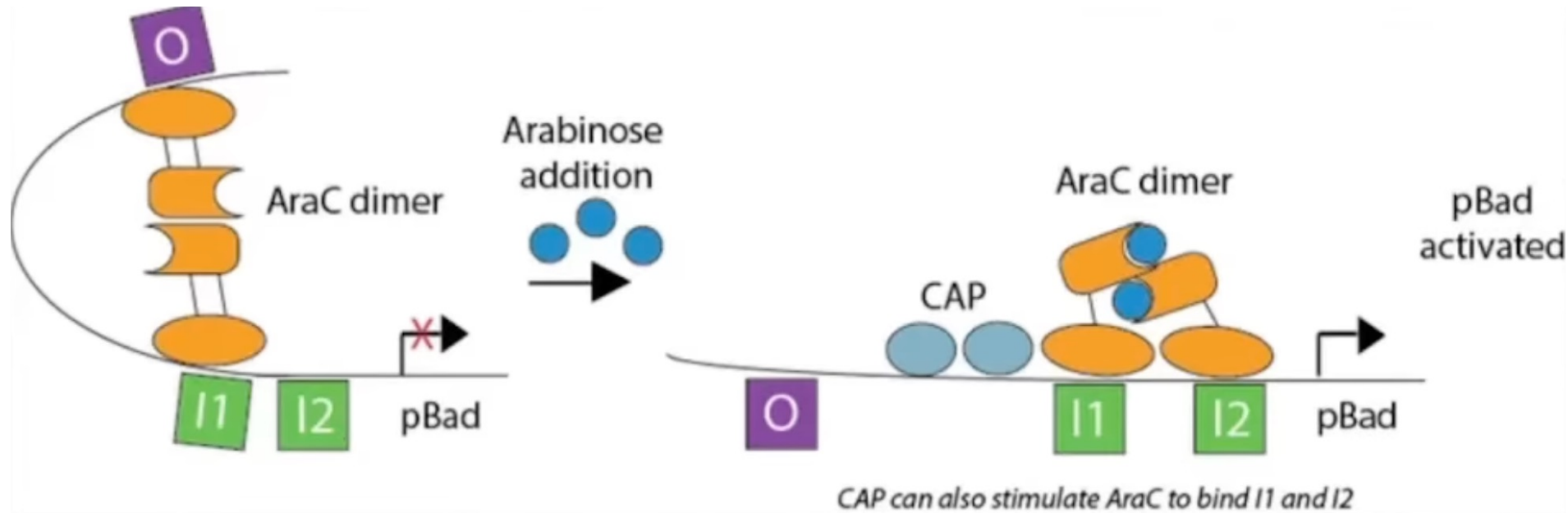
Gene regulation systems in synthetic biology



Isopropyl β -D-1-thiogalactopyranoside (IPTG)
Similar to allolactose



Gene regulation systems in synthetic biology



Arabinose (Ara) operon system

Notations reminder



Basic

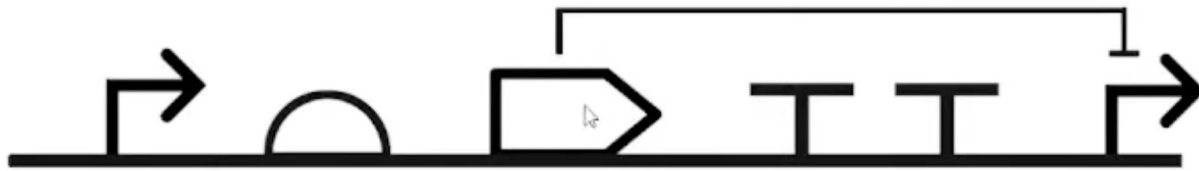


Activator



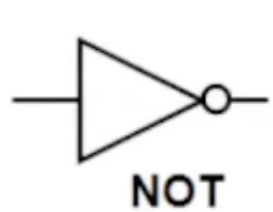
Bioreporter

Inverter

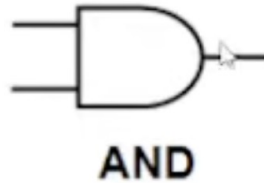


Lac repressor

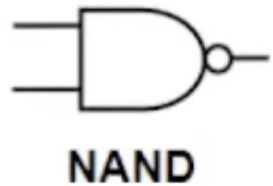
Boolean complete Logic gates of genetic circuits



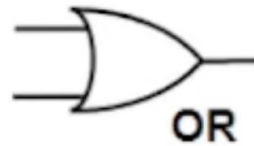
Input	Output
I	F
0	1
1	0



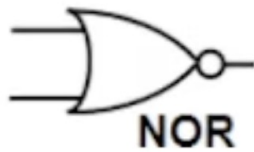
Inputs		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



Inputs		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0

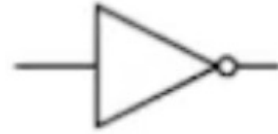


Inputs		Output
A	B	F
0	0	0
1	0	1
0	1	1
1	1	1



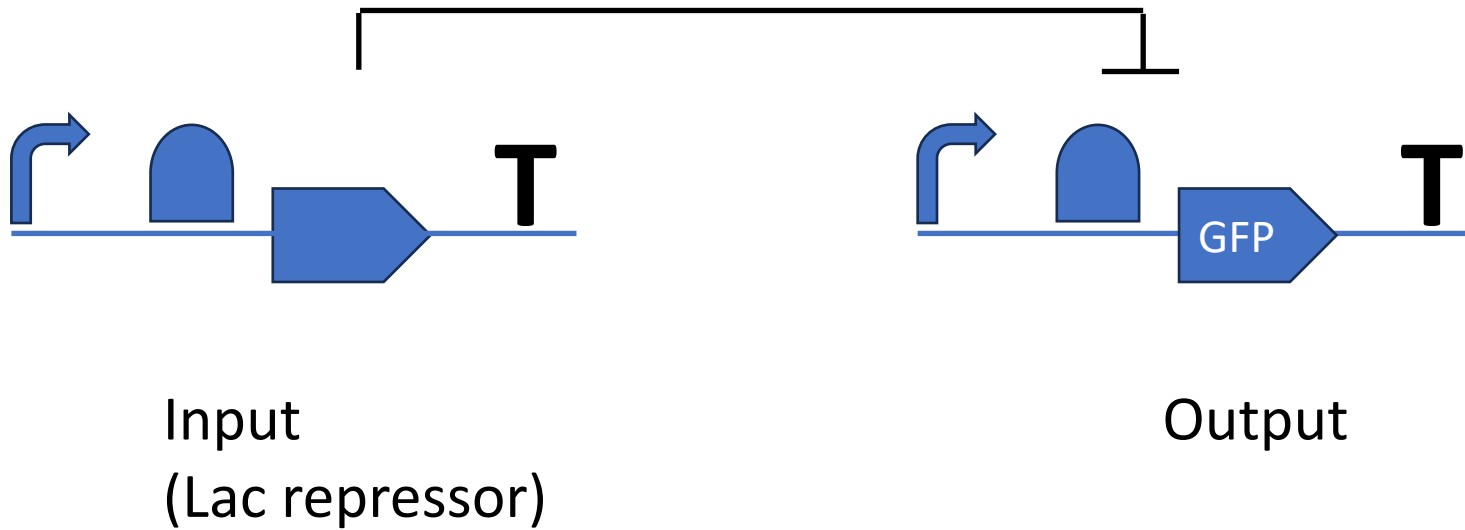
Inputs		Output
A	B	F
0	0	1
1	0	0
0	1	0
1	1	0

End gate



Truth Table

Input	Output
I	F
0	1
1	0



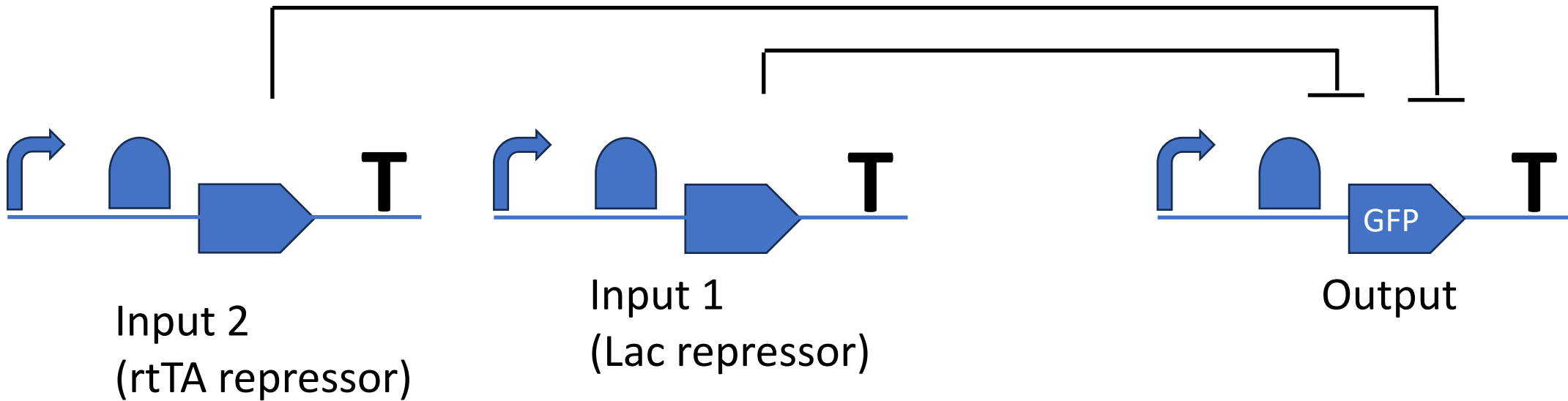
The inverter

AND Gate

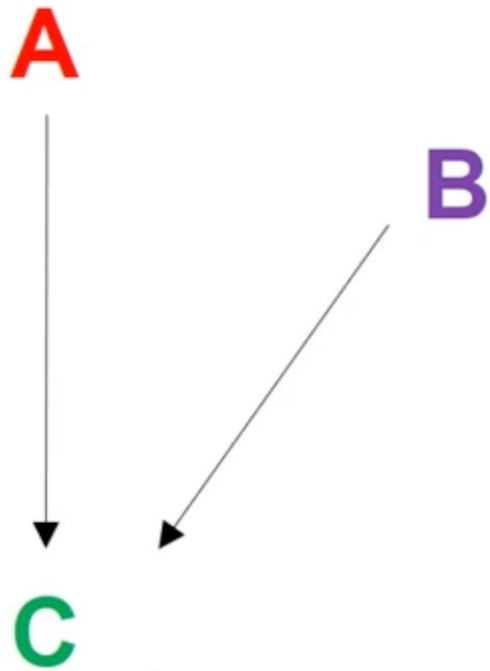


Truth Table

Input		Output
A	B	F
0	0	0
1	0	0
0	1	0
1	1	1



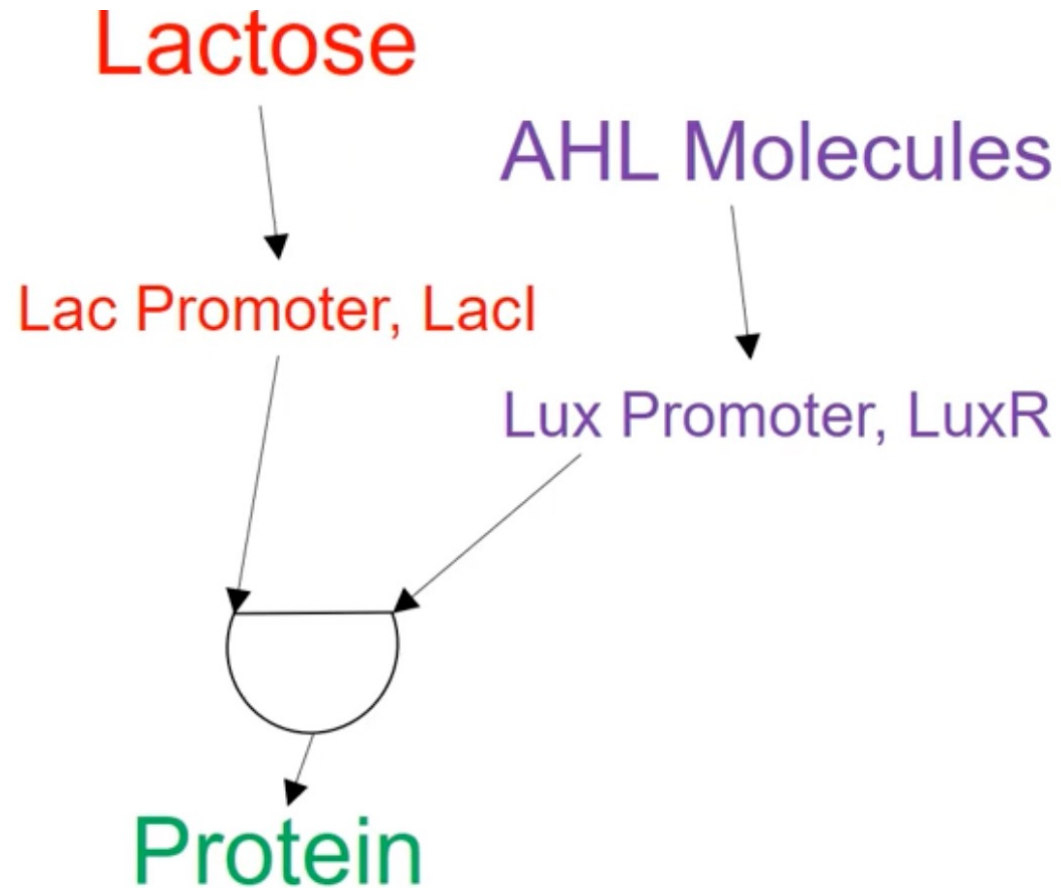
AND Gate: Two inputs



An **input (A)** and another **input (B)** is required to produce the **output (C)**

**Can be used to
integrated multiple
signals, prevent false
starts**

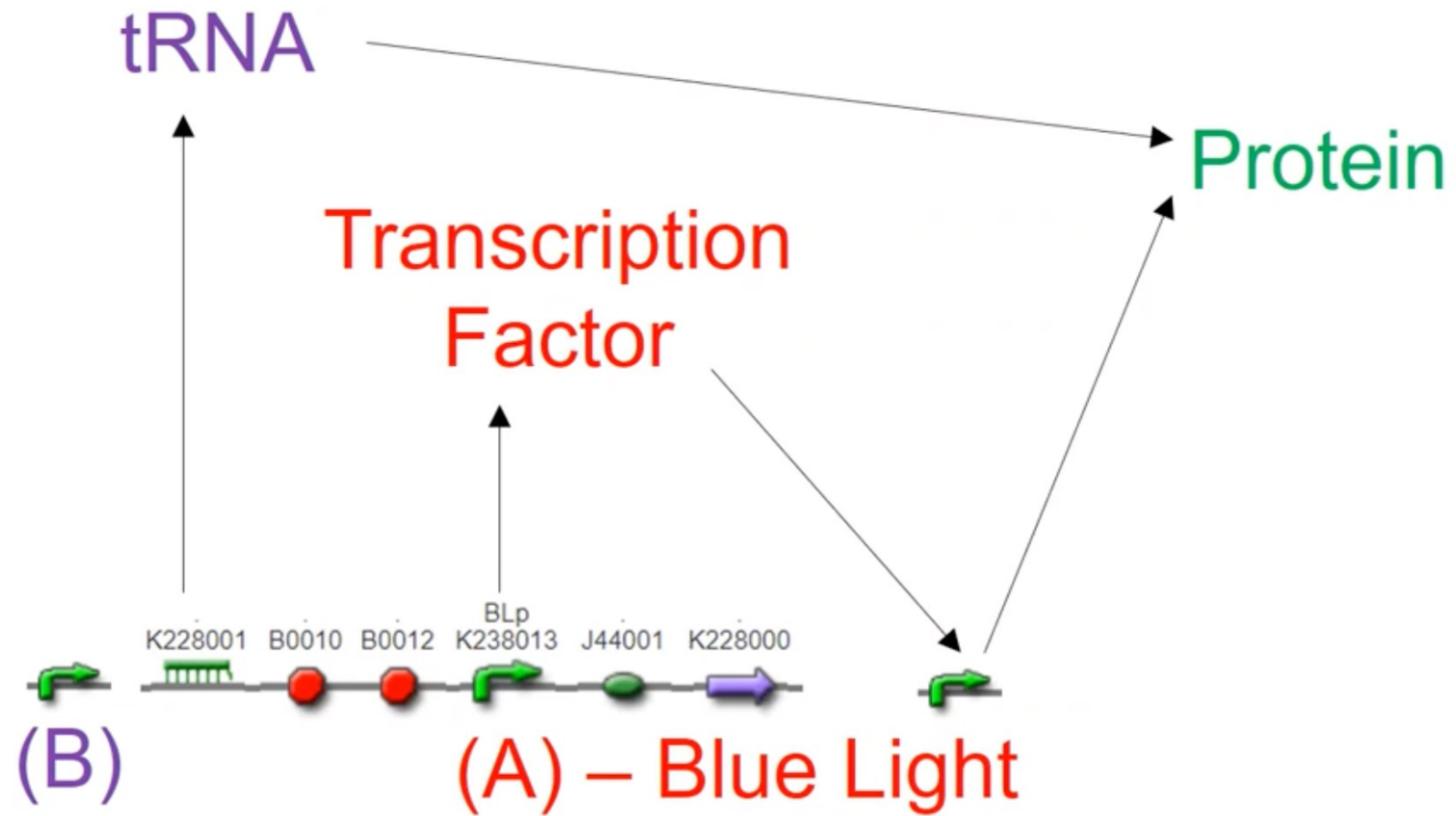
AND Gate: Two inputs



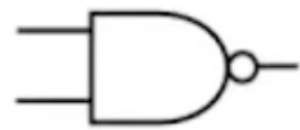
**Can be easily made
using a Hybrid Promoter:**
Contains multiple operator
/ repressor binding sites

**Only a true AND Gate if
both signals are required**

AND Gate: Two inputs

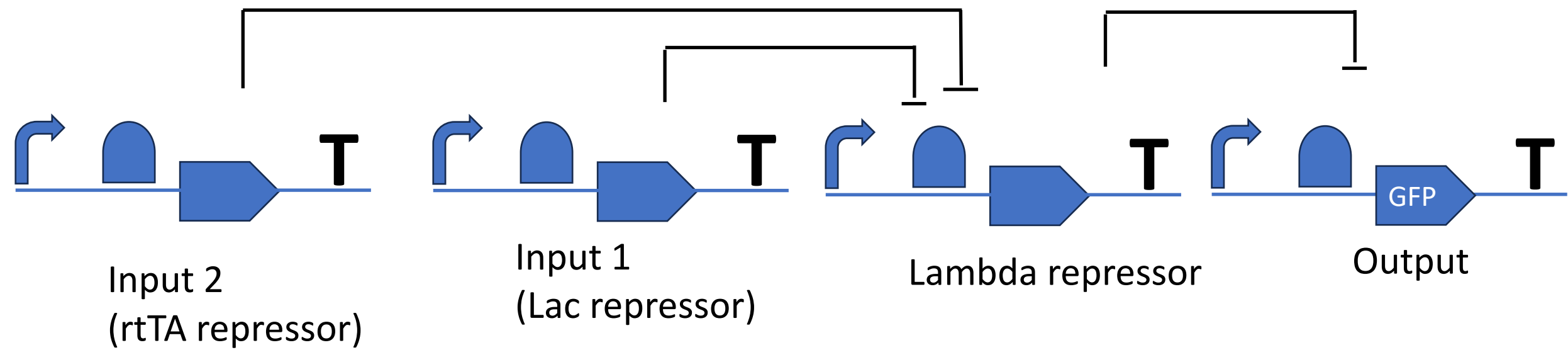


NAND Gate

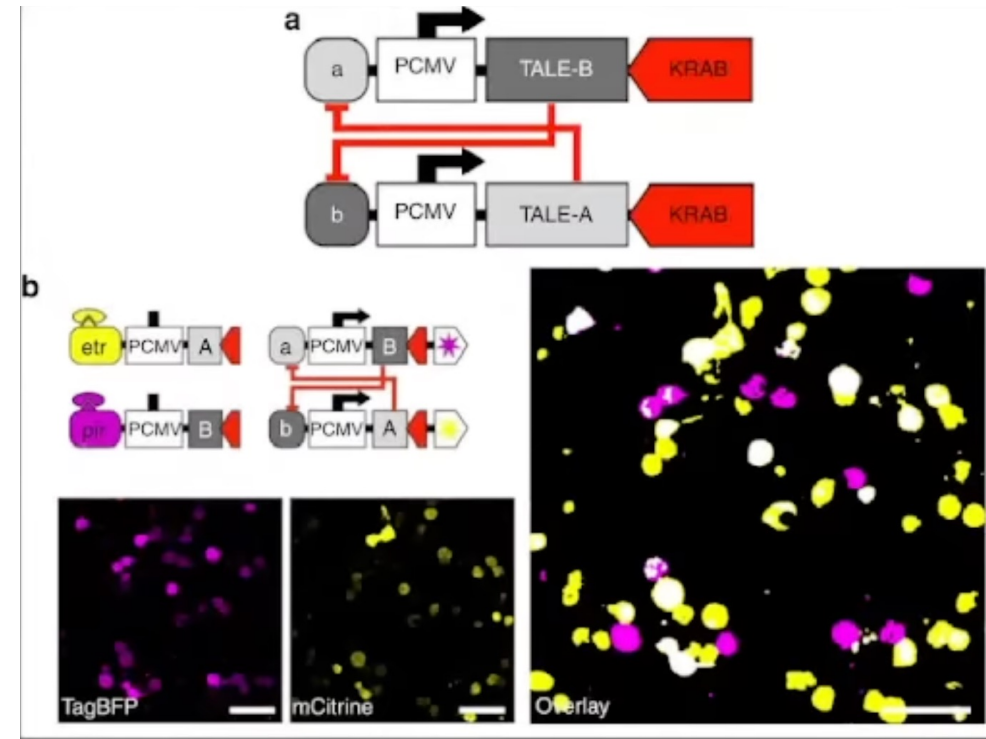
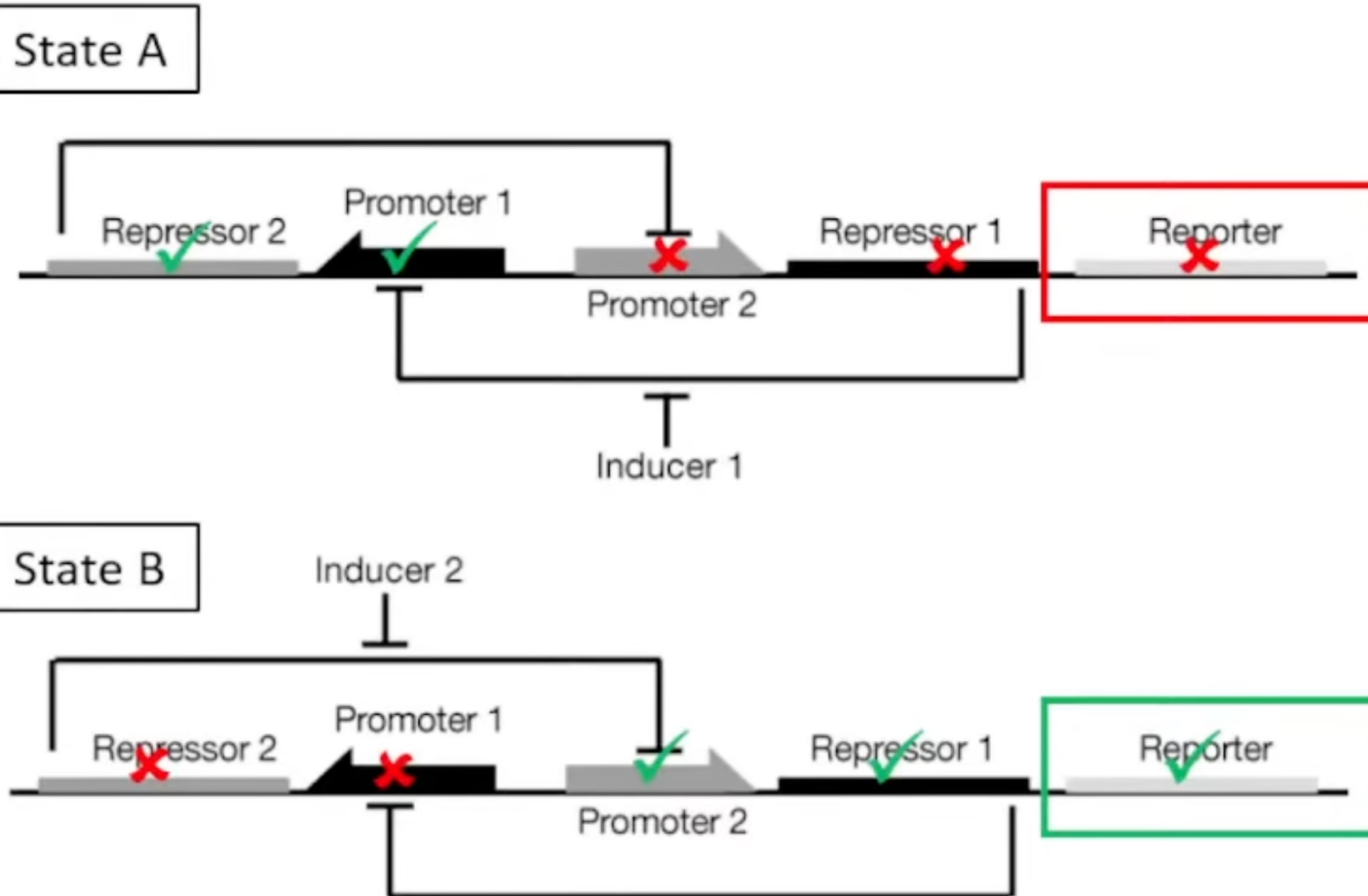


Truth Table

Input		Output
A	B	F
0	0	1
1	0	1
0	1	1
1	1	0



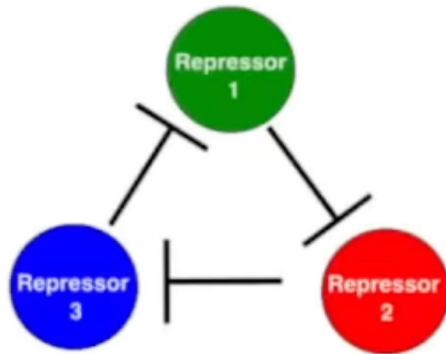
Bistable genetic switch



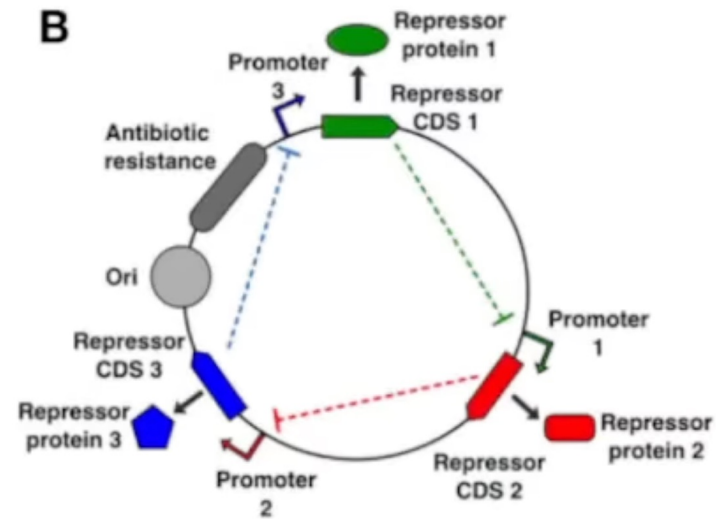
Tina Lebar, Nature Communications, 2014

Repressilator

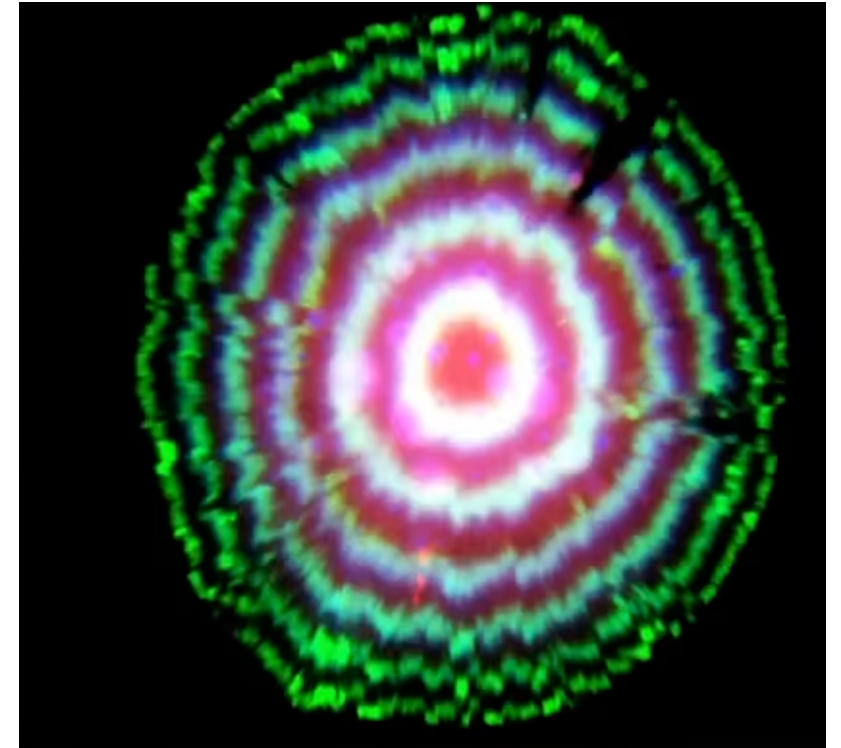
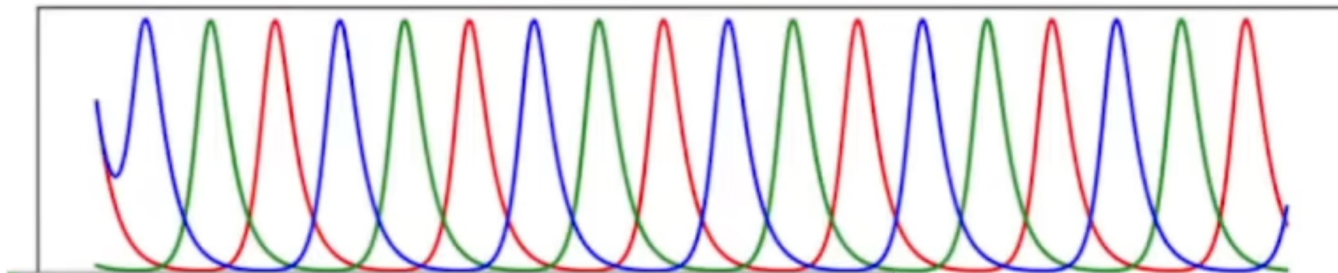
A

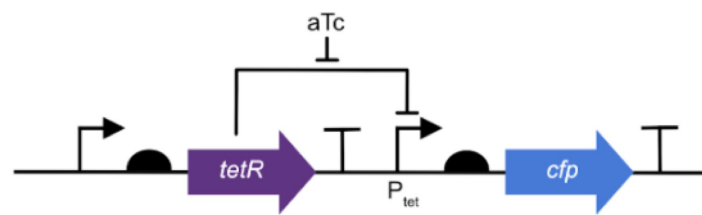


B

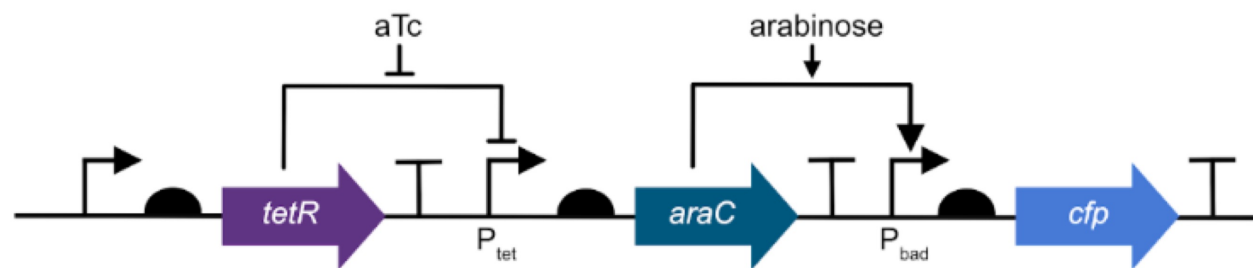


C





aTc	CFP
0	
1	



aTc	arabinose	CFP
0	0	
0	1	
1	0	
1	1	

Truth table

Logic gate

Create the circuit for Zinc biosensor

